

HeteroFormer: Prototype-Driven Unified Recommendation with Decoupled Semantic Optimization

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Abstract

Conversion rate (CVR) prediction demands jointly modeling long-range user behavior sequences and heterogeneous non-sequential feature under a unified representation. Existing architectures predominantly adopt a decoupled pipeline: sequences are first compressed into fixed-length vectors, which are then fused with dense features through token-mixing modules. This separation limits both the depth of sequential structure utilization and the flexibility of cross-field interaction. We present HeteroFormer, a unified architecture founded on a single principle: sequence-derived semantics should actively parameterize heterogeneous feature interactions, rather than being passively injected as compressed context. The core mechanism is a Dynamic Prototype Manifold—a set of learnable semantic anchors whose geometry is conditioned on user features via Cayley rotation and whose assignment is computed through entropy-regularized optimal transport (Sinkhorn). These prototype assignments serve as soft biases and gating signals within a cross-field interaction network, enabling user-specific sequential patterns to directly shape attention topologies and feed-forward transformations. To capture residual information without destabilizing the primary CVR objective, we introduce a generative semantic layer trained via Decoupled Semantic Optimization (DSO), where generative and discriminative gradients are architecturally isolated and their coupling is governed by an overfitting-aware MetaAligner. We further demonstrate that this pathway naturally induces semantic IDs—discrete, interpretable indices over the prototype manifold—that render the model’s reasoning transparent. While our primary goal is architectural innovation rather than leaderboard optimization, extensive training dynamics on the Tencent Ads Algorithm Competition dataset empirically validate that every module of DSO behaves according to its mechanistic design. An additional stability ablation confirms that the core principle of prototype-conditioned interaction remains effective even when the optimal-transport machinery is surgically replaced by lightweight alternatives, although the full generative calibration path is necessary for numerically stable inference. Code is available at: https://github.com/zhuzhuzhening/taac_2026.

1 Introduction

Industrial large-scale recommendation models (LRMs) face the fundamental challenge of jointly modeling long range user behavior sequences and heterogeneous nonsequential features under strict representational constraints. The prevailing paradigm follows a two-stage pipeline: (1) a sequence compressor such as DIN [1], BST [3], or LONGER [4] first aggregates variable-length behavior sequences into fixed-length query tokens; (2) a feature interaction module such as FM [6], DeepFM [7], or RankMixer [9] subsequently mixes these tokens with non-sequential features. While modular, this decoupled design inherently creates a representational bottleneck: sequential information is compressed once and for all before interaction, and cross-field interactions cannot feed back to refine sequence representations.

Recent works have attempted to bridge this gap from three distinct angles, yet none fully resolve the bottleneck:

- Alternating optimization (HyFormer [5]) iterates between query decoding and query boosting, but still treats sequence and interaction as distinct phases.
- Unified parameterization (UniMixer [11], MixFormer [12], TokenFormer [13]) unifies attention, TokenMixer, and FM under a single scaling framework, but does not explicitly address how sequential structure should inform the mixing pattern.
- Generative retrieval (OneRec, Kuaiformer) reformulates recommendation as next-token prediction, yet relies on beam search with prohibitive inference cost and lacks fine-grained feature interaction control.

We argue that the root cause of the bottleneck lies in a deeper assumption shared by all existing approaches: sequence semantics are treated as passive context. Whether compressed into a vector, decoded as a query, or generated as a token ID, the sequential signal is injected into the interaction module rather than parameterizing it. This leads to three concrete limitations:

1. *Passive sequence injection.* Compressed sequence vectors are concatenated or added to non-sequential tokens, acting as “dead” context rather than actively shaping interaction weights.
2. *Rigid feature interaction.* Token-mixing modules apply fixed permutation patterns or dense attention, lacking conditional adaptation based on user-specific sequential patterns.
3. *Under-utilized sequential structure.* High-level semantic concepts latent in behavior sequences (e.g., category-level preferences, temporal intent shifts) are not explicitly disentangled and leveraged as interpretable control signals.

To address these limitations, we propose HeteroFormer, whose contributions are summarized as follows:

- We introduce a Dynamic Prototype Manifold that represents behavior sequences as soft assignments over learnable semantic anchors. The prototype geometry is conditioned on user features through Cayley rotation, and assignment is formulated as entropy-regularized optimal transport via Langevin-Sinkhorn, providing differentiable, sparse, and interpretable sequence summarization.
- We propose Proto-Conditioned Cross-Field Interaction, where prototype assignments serve as soft biases in multi-head cross-field attention and as gating signals in FiLM-conditioned feed-forward networks. This tightly couples sequential semantics with heterogeneous feature interaction in a bidirectional, coevolutionary manner.
- We design a Generative Semantic Layer comprising a Diffusion Explainer (information-bottleneck residual encoder) and an Energy Calibrator (error-predictive scoring function). These modules capture complementary residual signals without interfering with the primary CVR gradient path. Importantly, we demonstrate that this generative pathway naturally induces semantic IDs—discrete, human-interpretable indices over the prototype manifold—that render the model’s sequential reasoning transparent.
- We present Decoupled Semantic Optimization (DSO), a training methodology that isolates generative and discriminative objectives through gradient detachment and governs their coupling via an overfitting-aware MetaAligner, ensuring stable co-training.

- We provide a critical analysis on why standard Transformer-based interaction is ill-suited for heterogeneous CVR features, and argue that prototype-driven assignment redefines explainability in modern recommendation systems by replacing opaque attention weights with semantically grounded mixture coefficients.
- We report a stability ablation that surgically replaces the SSM encoder and Sinkhorn prototype layer with lightweight, HyFormer-inspired alternatives (RoPE, SwiGLU, soft theme routing). The ablation confirms that the core principle of sequence-parameterized interaction remains trainable and effective even under severe resource constraints, while revealing that the full generative calibration path is necessary to prevent logit collapse and uncertainty degeneration.

2 Related Work

Sequence Modeling in CVR Prediction. Early approaches such as DIN [1] and DIEN [2] employ attention mechanisms to activate relevant behaviors with respect to target items. Transformer-based methods including BST [3] and LONGER [4] extend this to multilayer self-attention over long sequences. State-space models (SSMs) have recently gained traction for their linear complexity; structured variants such as S4 [14] and Mamba [15] offer principled long-range dependency modeling. HeteroFormer adopts a lightweight SSM cell for continuous-time sequence encoding, coupled with multi-scale aggregation to capture both short-term and long-term user intent.

Heterogeneous Feature Interaction. Factorization machines [6] and their deep variants (DeepFM [7], xDeepFM [8]) model second- and high-order feature interactions explicitly. TokenMixer [9] and its large-scale extension TokenMixer-Large [10] reformulate interaction as token-mixing via learned permutation matrices. UniMixer [11] unifies attention, TokenMixer, and FM under a generalized parameterized mixing framework with Sinkhorn-Knopp doubly-stochastic constraints. HyFormer [5] alternates between query decoding (sequence attention) and query boosting (token mixing). HeteroFormer differs by making the interaction pattern itself conditional on sequence-derived prototypes, rather than applying a fixed mixing operator.

Generative and Energy-Based Models in Recommendation. Diffusion models have been explored for sequential recommendation [16] and counterfactual explanation generation. Energy-based models (EBMs) provide a natural framework for uncertainty quantification and out-of-distribution detection. HeteroFormer draws inspiration from these directions but employs them in a residual, auxiliary capacity: the Diffusion Explainer compresses residual sequence information through an information bottleneck, while the Energy Calibrator predicts prediction error magnitude to enable dynamic logit calibration at inference time.

3 Methodology

3.1 Overview

The HeteroFormer architecture consists of four tightly integrated stages (Figure 1):

1. *Multi-View Encoding:* User, item, and dense features are encoded into shared semantic representations with task-specific projections.
2. *Continuous Sequence Modeling:* Variable-length behavior sequences are processed by SSM cells with continuous-time encoding, producing multi-scale summaries (short-term, long-term, static) that are fused across domains.

3. *Dynamic Prototype Manifold*: The fused sequence representation is mapped onto a user-conditioned prototype manifold via Cayley rotation and Langevin-Sinkhorn optimal transport, yielding sparse prototype assignments and a compact prototype representation.
4. *Proto-Conditioned Interaction & Prediction*: Prototype assignments bias cross-field attention and gate feed-forward transformations within the interaction network; a calibrated prediction head fuses static, prototype, and residual signals for the final CVR estimate.

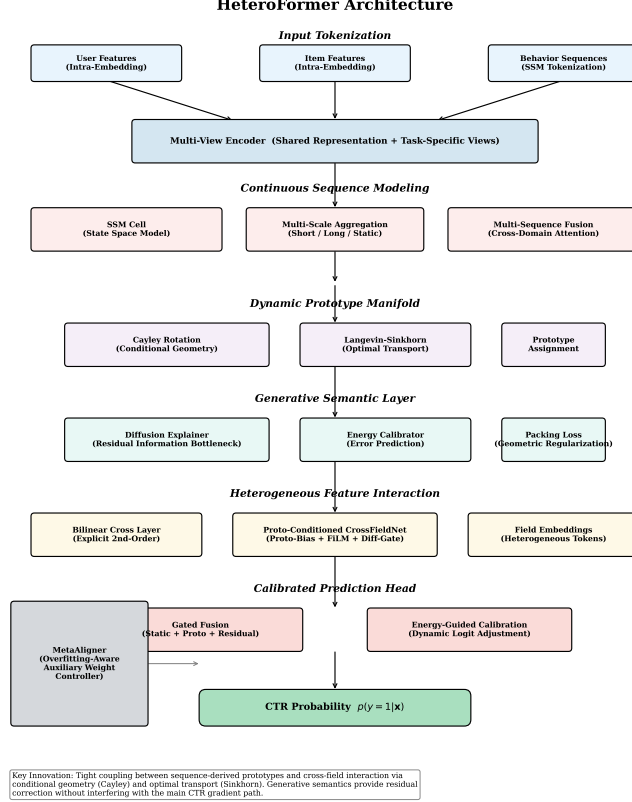


Figure 1: The HeteroFormer architecture. Sequence-derived prototypes from the Dynamic Prototype Manifold directly parameterize the Proto-Conditioned CrossFieldNet via soft biases and FiLM gating. The Generative Semantic Layer provides residual corrections through gradient-detached connections.

3.2 Multi-View Feature Encoding

Let $\mathbf{x}_u \in \mathbb{R}^{d_u}$, $\mathbf{x}_i \in \mathbb{R}^{d_i}$, $\mathbf{x}_{du}^{\text{dense}} \in \mathbb{R}^{d_{du}^{\text{dense}}}$, and $\mathbf{x}_i^{\text{dense}} \in \mathbb{R}^{d_i^{\text{dense}}}$ denote user ID features, item ID features, user dense features, and item dense features, respectively. Each ID feature j with vocabulary size V_j is embedded via a constrained embedding layer with max-norm projection:

$$\mathbf{e}_j = \text{Embed}_{\text{max-norm}}(x_j; V_j, d_e) = \frac{\mathbf{W}_j \boldsymbol{\sigma}(x_j)}{\|\mathbf{W}_j \boldsymbol{\sigma}(x_j)\|_2} \cdot \gamma, \quad (1)$$

where $\sigma(x_j)$ is the one-hot indicator, $\mathbf{W}_j \in \mathbb{R}^{d_e \times V_j}$, and γ is the maximum norm. For multi-slot features, we average slot embeddings. Dense features are linearly projected to dimension d_m .

User and item tokens are aggregated into field-wise representations $\mathbf{z}_u, \mathbf{z}_i \in \mathbb{R}^{d_m}$. A shared context vector is formed as $\mathbf{z}_{\text{ctx}} = \mathbf{z}_u + \mathbf{z}_i$, and task-specific views are generated via shallow MLPs for CVR, diffusion, and energy tasks. Formally:

$$\mathbf{z}_{\text{shared}} = \text{LayerNorm}(\mathbf{z}_u + \mathbf{z}_i + \mathbf{z}_{\text{ctx}}), \quad \mathbf{z}_{\text{view}}^{(t)} = \text{MLP}_{\text{view}}^{(t)}(\mathbf{z}_{\text{shared}}), \quad (2)$$

where $t \in \{\text{cvr}, \text{diff}, \text{energy}\}$.

3.3 Continuous Sequence Modeling

Behavior sequences are heterogeneous in both feature type and temporal granularity. For each sequence domain s , we embed discrete feature IDs into a concatenated representation $\mathbf{h}_{s,\ell}^{(0)} \in \mathbb{R}^{d_m}$ at position ℓ .

3.3.1 SSM Cell with Continuous-Time Discretization

We employ a structured state-space cell inspired by S4 [14]. Given input $\mathbf{x}_\ell \in \mathbb{R}^{d_m}$ and continuous time-step $\Delta_\ell > 0$, the state $\mathbf{h}_\ell \in \mathbb{R}^{d_s}$ evolves as:

$$\bar{\mathbf{A}}_\ell = \exp(\Delta_\ell \cdot \mathbf{A}), \quad \bar{\mathbf{B}}_\ell = \Delta_\ell \cdot \mathbf{B}, \quad \mathbf{h}_\ell = \bar{\mathbf{A}}_\ell \odot \mathbf{h}_{\ell-1} + \bar{\mathbf{B}}_\ell \mathbf{x}_\ell, \quad \mathbf{y}_\ell = \mathbf{C} \mathbf{h}_\ell + D \mathbf{x}_\ell, \quad (3)$$

where $\mathbf{A} \in \mathbb{R}^{d_s \times d_s}$ is diagonal with negative real entries initialized as $\mathbf{A}_{kk} = -\exp(k)$, $\mathbf{B} \in \mathbb{R}^{d_s \times d_m}$, $\mathbf{C} \in \mathbb{R}^{d_m \times d_s}$, and D is a learned skip scalar. In practice, we parameterize Δ_ℓ via a softplus projection of inter-event time differences:

$$\Delta_\ell = \text{softmax}(\text{MLP}(\tau_\ell - \tau_{\ell-1})). \quad (4)$$

3.3.2 Multi-Scale Aggregation and Cross-Domain Fusion

After L SSM layers, we derive three complementary summaries:

- *Short-term*: Exponentially weighted pooling over recent positions,

$$\mathbf{s}_{\text{short}} = \sum_{\ell} w_{\ell}^{(s)} \mathbf{y}_{\ell}, \quad \text{with } w_{\ell}^{(s)} \propto \exp(-\ell/\tau_s) \cdot \mathbb{I}[\ell < \ell_{\text{max}}]. \quad (5)$$

- *Long-term*: Uniform mean pooling over valid positions,

$$\mathbf{s}_{\text{long}} = \frac{1}{\ell_{\text{max}}} \sum_{\ell=1}^{\ell_{\text{max}}} \mathbf{y}_{\ell}. \quad (6)$$

- *Static*: A learned projection of mean, variance, and last-state statistics:

$$\mathbf{s}_{\text{static}} = \text{MLP}_{\text{stat}}([\bar{\mathbf{y}}; \boldsymbol{\sigma}_{\mathbf{y}}; \mathbf{y}_{\ell_{\text{max}}}]). \quad (7)$$

When multiple sequence domains exist (e.g., click, purchase, search), domain-specific summaries are projected and fused via cross-domain attention with a learnable query token:

$$\mathbf{z}_{\text{seq}} = \text{MultiHeadAttn}(\mathbf{q}_{\text{fuse}}, [\mathbf{s}_{\text{short}}^{(1)}, \dots, \mathbf{s}_{\text{short}}^{(s)}]), \quad (8)$$

where $\mathbf{q}_{\text{fuse}} \in \mathbb{R}^{d_m}$ is a learned global query.

3.4 Dynamic Prototype Manifold

The sequence representation $\mathbf{z}_{\text{seq}}$ is typically high-dimensional and noisy. We propose to map it onto a prototype manifold—a set of K learnable semantic anchors $\{\boldsymbol{\eta}_k\}_{k=1}^K \subset \mathbb{S}^{d_m-1}$ (unit hypersphere). The manifold geometry is conditioned on the user feature $\mathbf{z}_u$ through a Cayley rotation, and assignment is computed via entropy-regularized optimal transport.

3.4.1 User-Conditioned Cayley Rotation

To adapt prototypes to individual users while preserving the manifold structure, we apply a low-rank skew-symmetric update. Let $\mathbf{U}, \mathbf{V} \in \mathbb{R}^{d_m \times r}$ be learnable rank- r factors, and let $\mathbf{w} = \tanh(\text{MLP}(\mathbf{z}_u)) \in \mathbb{R}^r$ be user-specific scaling coefficients. The skew-symmetric matrix is:

$$\mathbf{G}(\mathbf{z}_u) = \mathbf{U} \text{diag}(\mathbf{w}) \mathbf{V}^\top - \mathbf{V} \text{diag}(\mathbf{w}) \mathbf{U}^\top. \quad (9)$$

The Cayley transform yields an orthogonal rotation matrix:

$$\mathbf{R}(\mathbf{z}_u) = (\mathbf{I} + \frac{1}{2}\mathbf{G})(\mathbf{I} - \frac{1}{2}\mathbf{G})^{-1}, \quad (10)$$

and the user-conditioned prototypes are:

$$\boldsymbol{\mu}_k(\mathbf{z}_u) = \mathbf{R}(\mathbf{z}_u) \boldsymbol{\eta}_k, \quad \boldsymbol{\mu}_k \in \mathbb{S}^{d_m-1}. \quad (11)$$

This ensures that prototype norms are preserved and the transformation is invertible, avoiding representational collapse.

3.4.2 Concentration-Modulated Similarity

The affinity between the sequence representation and each prototype is modulated by a user- and time-dependent concentration parameter κ . We parameterize:

$$\kappa(\mathbf{z}_u, \Delta t) = \text{softmax}(\kappa_{\text{base}} + \text{MLP}_\kappa(\mathbf{z}_u)) \cdot \exp(-\lambda_\kappa \Delta t / 86400) + \kappa_{\text{min}}, \quad (12)$$

where Δt is the time span of the sequence (in seconds), and λ_κ is a learned decay rate. This captures the intuition that older sequences should yield more diffuse prototype assignments.

The unnormalized log-probability (logit) of assignment is:

$$\ell_k = \kappa \cdot \tanh(\langle \bar{z}_{\text{seq}}, \boldsymbol{\mu}_k(\mathbf{z}_u) \rangle), \quad (13)$$

where $\bar{z}_{\text{seq}}$ is ℓ_2 -normalized. The tanh compresses extreme similarities to prevent sharp assignment collapse during early training.

3.4.3 Langevin-Sinkhorn Assignment

To obtain sparse, differentiable, and doubly-stochastic assignment weights, we frame prototype assignment as entropy-regularized optimal transport between a sequence “mass” and prototype “bins.” Given cost matrix $\mathbf{C} = -\ell \mathbf{1}^\top$, the optimal coupling $\boldsymbol{\Pi}^*$ solves:

$$\boldsymbol{\Pi}^* = \arg \min_{\boldsymbol{\Pi} \in \mathcal{U}(r, c)} \langle \mathbf{C}, \boldsymbol{\Pi} \rangle - \epsilon H(\boldsymbol{\Pi}), \quad (14)$$

where $\mathcal{U}(r, c)$ is the set of matrices with prescribed row and column marginals, $H(\boldsymbol{\Pi}) = -\sum_{i,j} \Pi_{ij} \log \Pi_{ij}$ is the entropy, and $\epsilon > 0$ is the regularization coefficient. We approximate $\boldsymbol{\Pi}^*$ via Sinkhorn-Knopp iteration:

$$\mathbf{u}^{(t+1)} = \frac{\mathbf{r}}{\mathbf{K} \mathbf{v}^{(t)}}, \quad \mathbf{v}^{(t+1)} = \frac{\mathbf{c}}{\mathbf{K}^\top \mathbf{u}^{(t+1)}}, \quad (15)$$

where $\mathbf{K} = \exp(-\mathbf{C}/\epsilon)$. During training, we inject Langevin noise into the dual variables for implicit regularization:

$$\mathbf{u}^{(t+1)} \leftarrow \mathbf{u}^{(t+1)} + \sigma_{\text{noise}} \cdot \boldsymbol{\xi}, \quad \boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (16)$$

The final assignment weights are $\pi = \text{diag}(\mathbf{u})\mathbf{K}\text{diag}(\mathbf{v})$ and the prototype representation is:

$$\mathbf{p} = \sum_{k=1}^K \pi_k \boldsymbol{\mu}_k(\mathbf{z}_u), \quad \bar{\mathbf{p}} = \text{normalize}(\mathbf{p}). \quad (17)$$

3.4.4 Geometric Packing Regularization

To encourage prototypes to tile the semantic space uniformly, we impose a packing loss on the global anchors:

$$\mathcal{L}_{\text{pack}} = \sum_{k \neq k'} \text{ReLU}(|\langle \boldsymbol{\eta}_k, \boldsymbol{\eta}_{k'} \rangle| - \tau_{\text{coh}})^2 + \lambda_{\text{rot}} \left(\|\mathbf{U}^\top \mathbf{U} - \mathbf{I}\|_F^2 + \|\mathbf{V}^\top \mathbf{V} - \mathbf{I}\|_F^2 \right), \quad (18)$$

where τ_{coh} is a coherence threshold and λ_{rot} penalizes deviation from orthogonality in the Cayley factors.

3.5 Generative Semantic Layer: Inducing Semantic IDs for Interpretability

While the prototype manifold provides a compressed semantic summary, behavior sequences contain rich residual information (e.g., fine-grained temporal dynamics, out-of-distribution interactions) that may improve prediction if harnessed without destabilizing the main CVR objective. We introduce two complementary generative modules. Crucially, we argue that the generative pathway does not merely serve as an auxiliary loss vehicle; it induces a semantic ID system over the prototype manifold that renders the model’s sequential reasoning transparent and auditable.

3.5.1 Semantic IDs as Prototype Assignments

In standard embedding-based recommendation, the “ID” of a user’s sequential preference is implicitly encoded as a continuous vector in $\mathbb{R}^{d_m}$, which is opaque to human inspection. HeteroFormer replaces this implicit encoding with an explicit soft assignment $\pi \in \Delta^{K-1}$ over a finite vocabulary of K prototypes. Each prototype η_k can be interpreted as a latent semantic concept (e.g., “price-sensitive browsing,” “impulse purchasing,” “research-heavy comparison”). The assignment vector π therefore functions as a semantic ID: a probabilistic membership vector that tells us, in human-interpretable terms, which behavioral modes are active for a given user.

Formally, because the Sinkhorn iteration projects onto the probability simplex, π satisfies:

$$\pi_k \geq 0, \quad \sum_{k=1}^K \pi_k = 1, \quad \pi_k \propto \exp\left(\frac{\ell_k + u_k + v_k}{\epsilon}\right), \quad (19)$$

where u_k, v_k are the dual potentials. The entropy $H(\pi)$ measures assignment uncertainty; low entropy indicates a crisp semantic ID (e.g., $\pi \approx e_3$), while high entropy indicates mixed or ambiguous behavior. This is structurally analogous to the “soft token” representations in vision transformers, but here the tokens have explicit semantic grounding through the packing loss, which forces prototypes to cover distinct regions of the behavioral manifold.

3.5.2 Diffusion Explainer: Residual Information Bottleneck

The Diffusion Explainer encodes the residual between the sequence representation and its prototype summary through an information bottleneck. Let $\mathbf{z}_{\text{seq}}$, π , and $\{\mu_k\}$ be as defined above. The input to the explainer is:

$$\mathbf{x}_{\text{diff}} = [\mathbf{z}_{\text{seq}}; \sum_k \pi_k \mu_k; \pi] \in \mathbb{R}^{2d_m + K}. \quad (20)$$

An encoder $\text{Enc}_{\text{diff}} : \mathbb{R}^{2d_m + K} \rightarrow \mathbb{R}^{d_b}$ with $d_b < d_m$ produces a bottleneck representation $\mathbf{h}_{\text{diff}}$. The explanation vector is decoded as:

$$\delta = \text{Dec}_{\text{diff}}(\mathbf{h}_{\text{diff}}) \in \mathbb{R}^{4d_m}, \quad (21)$$

where the expanded dimension $4d_m$ matches the concatenated field representation in the interaction network (Section 3.7).

To ensure δ captures genuinely complementary information, we enforce orthogonality with respect to the CVR representation $\mathbf{r}_{\text{cvr}}$:

$$\delta_{\perp} = \delta - \frac{\langle \delta, \mathbf{r}_{\text{cvr}} \rangle}{\|\mathbf{r}_{\text{cvr}}\|_2^2} \mathbf{r}_{\text{cvr}}, \quad (22)$$

and define the alignment loss:

$$\mathcal{L}_{\text{align}} = \|\langle \delta, \mathbf{r}_{\text{cvr}} \rangle\|_2^2 - \lambda_{\text{div}} \|\delta_{\perp}\|_2^2. \quad (23)$$

The first term suppresses redundant information; the second term encourages diversity in the residual signal. Additionally, a boundary-aware residual target encourages larger explanations for uncertain predictions:

$$\mathcal{L}_{\text{res}} = \|\delta\|_2^2 - \mathbb{I}[0.3 < \sigma(\ell_{\text{base}}) < 0.7] \cdot 0.5 \|\delta\|_2^2. \quad (24)$$

The information bottleneck enforces that $\mathbf{h}_{\text{diff}}$ discards all information already captured by π , thereby isolating novel sequential signals. This is formally related to the Information Bottleneck principle [17]:

$$\min I(\mathbf{h}_{\text{diff}}; \mathbf{x}_{\text{diff}}) - \beta I(\mathbf{h}_{\text{diff}}; y), \quad (25)$$

where β trades off compression and predictive relevance. In our setting, because $\mathbf{x}_{\text{diff}}$ already contains π , the bottleneck is encouraged to encode only the residual $R = \mathbf{z}_{\text{seq}} - \sum_k \pi_k \mu_k$, which represents the portion of user behavior not explained by known semantic prototypes. When this residual is large, it signals the emergence of a novel behavioral pattern—a natural trigger for prototype manifold expansion or human-in-the-loop auditing.

3.5.3 Energy Calibrator: Uncertainty as Explainability

The Energy Calibrator predicts the expected prediction error as a function of prototype assignments and nonsequential features. Let ℓ_{base} be the logit from the base CVR path (without residual correction). The energy score is:

$$e = \text{MLP}_{\text{energy}}(\{\pi; \mathbf{z}_u; \mathbf{z}_i; |\ell_{\text{base}}|\}) \in \mathbb{R}_{>0}, \quad (26)$$

constrained to positive values via a final softplus activation. During training, the energy target is derived from the BCE error:

$$e_{\text{target}} = \text{clamp}(5 \cdot (\sigma(\ell_{\text{base}}), y), 0.01, 3.0), \quad (27)$$

and the calibration loss is $\mathcal{L}_{\text{energy}} = (e - e_{\text{target}})^2$. At inference, high energy indicates low confidence, enabling dynamic logit attenuation.

From an explainability perspective, the energy score e serves as a model-aware confidence metric that is directly grounded in the semantic ID π . When e is high, we can inspect π to determine which prototype combination triggered the uncertainty. For instance, a user with high membership in both “budget-conscious” and “luxury-seeking” prototypes (conflicting semantics) will naturally yield high energy, alerting the system to a potentially inconsistent user profile.

3.6 Why Not Standard Transformer? Rethinking Interaction for Heterogeneous CVR Features

A natural question arises: why does HeteroFormer not simply stack standard Transformer layers over the concatenated field tokens, as done in BERT4Rec [20] or many recent CVR models? We argue that standard self-attention is structurally mismatched to the heterogeneous, sparse, and semantically imbalanced nature of CVR features. Our design of the Proto-Conditioned CrossFieldNet is a deliberate departure motivated by three fundamental observations.

3.6.1 Observation 1: Feature Heterogeneity Violates Token Homogeneity

In NLP or vision, all tokens belong to the same modality and approximately share the same statistical properties (vocabulary distribution, scale, sparsity). In CVR prediction, the four canonical fields—user, item, context, sequence—are profoundly heterogeneous:

- User features are extremely sparse (one-hot IDs), high-dimensional, and static within a session.
- Item features are dense catalog attributes with strong structural correlations (category $\rightarrow$ brand $\rightarrow$ price).
- Context features are ephemeral (time, location, device) and often have heavy-tailed distributions.
- Sequence features are variable-length, temporally ordered, and semantically multi-scale.

When these fields are flattened into a single token sequence and fed to standard self-attention, the query-key dot product $\langle \mathbf{q}_i, \mathbf{k}_j \rangle$ conflates four incompatible inner-product geometries. The attention weight from a userID token to an item-price token is mathematically defined but semantically meaningless, because the two tokens live on different statistical manifolds. Formally, if each field m has its own metric tensor $\mathbf{G}_m$, the standard attention assumes a shared Euclidean metric $\mathbf{I}$, which induces a distorted similarity measure:

$$\text{sim}_{\text{naive}}(i, j) = \mathbf{f}_i^\top \mathbf{f}_j, \quad \text{vs.} \quad \text{sim}_{\text{true}}(i, j) = \mathbf{f}_i^\top \mathbf{G}_{m(i), m(j)} \mathbf{f}_j. \quad (28)$$

HeteroFormer addresses this by maintaining field-specific embeddings and applying cross-field attention only through controlled, prototype-biased mechanisms, rather than allowing every token to attend to every other token indiscriminately.

3.6.2 Observation 2: Permutation Invariance is a Liability, Not a Virtue

Self-attention is permutation-invariant (up to positional encoding). In CVR features, however, field order carries strong semantic meaning: user features always precede item features in the

logical inference chain “who is interacting with what.” A permutation-invariant mixer destroys this causal structure. Our CrossFieldNet explicitly stacks field tokens as $[\mathbf{f}_{\text{user}}, \mathbf{f}_{\text{item}}, \mathbf{f}_{\text{ctx}}, \mathbf{f}_{\text{seq}}]$ and applies attention within this structured arrangement, preserving the asymmetric nature of user-item interaction. The bilinear cross layer (Section 3.7) further enforces pairwise interaction in a fixed, interpretable topology.

3.6.3 Observation 3: Quadratic Complexity is Prohibitively Expensive at Industrial Scale

Even if one ignores the semantic mismatch, standard Transformer attention has $O(M^2d)$ complexity, where M is the number of tokens. In industrial CVR models, M can reach thousands when multi-value categorical features are exploded into individual tokens. By contrast, HeteroFormer treats each field as a single token $M = 4$, reducing attention to $O(16d)$, while delegating fine-grained intra-field modeling to the SSM and prototype modules. The prototype assignment itself is $O(K^2)$ per sample (for Sinkhorn), which is negligible for $K \ll M$ (e.g., $K = 128$).

3.7 Proto-Conditioned Heterogeneous Feature Interaction

Traditional feature interaction modules operate on a fixed set of tokens (user, item, context, sequence) with parameter-agnostic mixing weights. HeteroFormer makes the interaction conditional on prototype assignments through two mechanisms: soft attention biasing and FiLM gating.

3.7.1 Bilinear Cross Layer

Before the main interaction network, we inject explicit second-order interactions via a lightweight bilinear layer. Given field tokens $\{f_m\}_{m=1}^M$ with $M = 4$ (user, item, context, sequence), the interaction between fields i and j is:

$$f'_i = f_i + \alpha \sum_{j>i} (f_i \odot w_{ij} \odot f_j), \quad f'_j = f_j + \alpha \sum_{j>i} (f_i \odot w_{ij} \odot f_j), \quad (29)$$

where $w_{ij} \in \mathbb{R}^{d_m}$ is a learnable cross-weight and α is a small ratio (e.g., 0.1). This provides cheap, interpretable pairwise interaction without increasing token count.

3.7.2 Proto-Conditioned CrossFieldNet

The core interaction stack consists of N layers, each comprising multi-head self-attention, prototype-biased cross-attention, and a FiLM-conditioned feed-forward network.

Prototype-Biased Self-Attention. Standard self-attention over field tokens $\mathbf{F} \in \mathbb{R}^{M \times d_m}$ computes:

$$\text{Attn}(\mathbf{F}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_h}} \right) \mathbf{V}. \quad (30)$$

We augment the attention output with a prototype-derived bias:

$$\tilde{\mathbf{F}} = \text{LayerNorm}(\mathbf{F} + \text{Attn}(\mathbf{F}) + \beta \cdot \text{MLP}_{\text{bias}}(\boldsymbol{\pi})), \quad (31)$$

where $\text{MLP}_{\text{bias}} : \mathbb{R}^K \rightarrow \mathbb{R}^{d_m}$ expands the assignment vector to token dimension, and β controls the injection strength. This allows sequential semantics to directly modulate which fields attend to each other. Notably, because π is sparse and semantically grounded, we can read off the attention

modulation: if π_k is large for a “price-sensitive” prototype, the bias term will amplify attention between user-income and item-price tokens.

FiLM Conditioning with Diffusion Gate. The feed-forward network is conditioned on the sequence token f_{seq} via Feature-wise Linear Modulation (FiLM):

$$\text{FiLM}(\mathbf{f}; \mathbf{f}_{\text{seq}}) = \gamma(\mathbf{f}_{\text{seq}}) \odot \mathbf{f} + \beta(\mathbf{f}_{\text{seq}}), \quad (32)$$

where γ, β are sigmoid- and linear-projections of f_{seq} respectively. Furthermore, the diffusion explainer output δ gates the FFN output:

$$f'_{\text{ffn}} = \text{FFN}(\hat{f}) \odot \sigma(\text{MLP}_{\text{gate}}(\hat{\delta})), \quad (33)$$

where σ is the sigmoid. The gating mechanism ensures that generative semantics selectively amplify or suppress channel activations without adding uncontrolled variance.

3.8 Calibrated Prediction Head

The prediction head fuses three information streams: (1) the static interaction representation $\mathbf{r}_{\text{static}} \in \mathbb{R}^{d_m}$ from the cross-field network; (2) the prototype representation $\mathbf{p} \in \mathbb{R}^{d_m}$; and (3) the residual explanation $\delta \in \mathbb{R}^{4d_m}$.

3.8.1 Gated Fusion

A three-way soft-attention gate computes mixing coefficients:

$$[g_{\text{stat}}, g_{\text{proto}}, g_{\text{res}}] = \text{softmax}(\text{MLP}_{\text{gate}}([\mathbf{z}_u; \mathbf{z}_i; \boldsymbol{\pi}; \boldsymbol{\epsilon}])), \quad (34)$$

where ϵ is the energy score. The fused representation is:

$$\mathbf{r}_{\text{fused}} = g_{\text{stat}} \cdot \mathbf{r}_{\text{static}} + g_{\text{proto}} \cdot (\mathbf{p} + \text{MLP}_{\text{boost}}(\mathbf{p})) + g_{\text{res}} \cdot \text{MLP}_{\text{res}}(\boldsymbol{\delta}) \cdot \sigma(\epsilon - 1), \quad (35)$$

where $\sigma(\epsilon - 1)$ attenuates the residual contribution when the energy score is low (high confidence), preventing over-reliance on auxiliary signals for easy examples.

3.8.2 Dynamic Logit Calibration

The final logit is computed as $\ell = \text{MLP}_{\text{out}}(\mathbf{r}_{\text{fused}}) \in \mathbb{R}$ followed by temperature scaling:

$$\hat{\ell} = \frac{\ell}{\tau} + b, \quad \tau = \exp(\tau_{\log}) \in [0.1, 5.0], \quad (36)$$

where $\tau_{\log}$ and b are learned. At inference, the energy score $\mathcal{E}$ provides an additional calibration: if $e > \tau_{\text{energy}}$ the model outputs a conservative prediction near the prior CVR, deferring uncertain decisions to downstream rankers.

3.9 Decoupled Semantic Optimization

Training HeteroFormer requires balancing three competing objectives: (1) the primary CVR classification loss; (2) the generative semantic losses (alignment, energy, packing); and (3) the geometric regularization of the prototype manifold. Naively summing these objectives leads to gradient interference and instability. We propose Decoupled Semantic Optimization (DSO), illustrated in Figure 2.

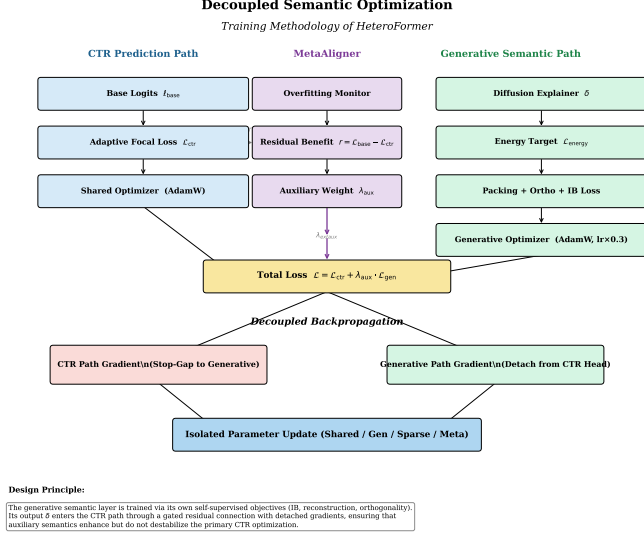


Figure 2: Decoupled Semantic Optimization. The CVR path and Generative Semantic path share forward computation but receive isolated gradients. MetaAligner modulates the coupling weight λ_{aux} based on residual benefit and overfitting gap.

3.9.1 Objective Decomposition

The total loss is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cvr}} + \lambda_{\text{aux}} \cdot \mathcal{L}_{\text{gen}}, \quad (37)$$

where the CVR loss is adaptive focal loss:

$$\mathcal{L}_{\text{cvr}} = \mathbb{E}_{(\mathbf{x}, y)} \left[\alpha_t (1 - p_t)^\gamma \cdot (\sigma(\hat{\ell}), y) \right] + \lambda_t \frac{|\gamma - \gamma_{\min}|}{\gamma_{\max} - \gamma_{\min}}, \quad (38)$$

with $p_t = y\sigma(\hat{\ell}) + (1 - y)(1 - \sigma(\hat{\ell}))$, $\alpha_t = y\alpha_+ + (1 - y)\alpha_-$, and γ is learned via a bounded sigmoid parameter. The generative loss aggregates:

$$\mathcal{L}_{\text{gen}} = \mathcal{L}_{\text{align}} + \lambda_{\text{energy}} \mathcal{L}_{\text{energy}} + \lambda_{\text{pack}} \mathcal{L}_{\text{pack}}. \quad (39)$$

3.9.2 Gradient Isolation via Detachment

The key to stable co-training is architectural gradient isolation rather than multi-step optimization:

- The base logit ℓ_{base} (computed without δ) is detached before entering the Diffusion Explainer, ensuring that generative gradients do not backpropagate through the CVR head.
- The diffusion output δ is detached from the generative path when computing $\mathcal{L}_{\text{cvr}}$; only the gating MLP in the interaction network receives CVR gradients.
- Prototype parameters $\{\eta_k, \mathbf{U}, \mathbf{V}\}$ participate in both paths but are updated by the shared optimizer, with gradient norms monitored independently.

3.9.3 MetaAligner: Overfitting-Aware Auxiliary Control

The coupling weight λ_{aux} is not a hyperparameter but a dynamic variable controlled by MetaAligner, which operates as an overfitting monitor without requiring a separate validation set at each step. Define the residual benefit:

$$r = \mathcal{L}_{\text{cvr}}^{(\text{base})} - \mathcal{L}_{\text{cvr}}^{(\text{final})}, \quad (40)$$

where $\mathcal{L}_{\text{cvr}}^{(\text{base})}$ is the loss without generative residuals and $\mathcal{L}_{\text{cvr}}^{(\text{final})}$ is the loss with residuals. A positive r indicates that generative semantics improve training loss. MetaAligner maintains an EMA of residual benefit, $\bar{r}$ and an EMA of the train-validation AUC gap, $\bar{g}$. The target auxiliary weight is:

$$\lambda_{\text{target}} = \text{clip} \left(\lambda_0 \cdot \frac{\bar{r}}{1 + \bar{r}}, 0.01, 0.5 \right), \quad (41)$$

clipped by the overfitting gap:

$$\lambda_{\text{target}} \leftarrow \lambda_{\text{target}} \cdot \max(0.1, 1 - 10(\bar{g} - 0.03)_+). \quad (42)$$

The final weight is smoothed via adaptive EMA with fast ($\rho_f = 0.8$) and slow ($\rho_s = 0.95$) decay rates selected by the magnitude of the update.

3.9.4 Isolated Parameter Update

Parameters are partitioned into four semantic groups: shared (encoder, interaction, head), gen (prototype geometry, diffusion, energy), sparse (embedding tables with Adagrad), and meta (MetaAligner and temperature). Each group maintains its own optimizer state. At each step:

1. Compute $\mathcal{L}_{\text{total}}$ and call backward() once.
2. Check gradient health; if generative gradients exceed a threshold, mask them.
3. Step shared and sparse optimizers jointly; step gen and meta independently.

This avoids the computational overhead of multiple backward passes while preserving semantic isolation through architectural detachment.

4 Experiments and Training Dynamics

The primary goal of this section is not to chase leaderboard rankings, but to provide a mechanistic validation of HeteroFormer’s core design hypothesis: that Decoupled Semantic Optimization can stably co-train a multi-objective architecture, and that each module behaves according to its intended control logic. We report training dynamics from the full HeteroFormer implementation, developed during the 2026 Tencent Ads Algorithm Competition, where the model was trained on a single GPU with full TensorBoard instrumentation.

4.1 Experimental Setup

Dataset. We use the TencentGR dataset provided by the 2026 Tencent Ads Algorithm Competition, which contains industrial-scale user behavior sequences and heterogeneous multi-field features. Due to the competition timeline, we report results from a single training run with extensive diagnostic logging rather than from a hyperparameter sweep.

Evaluation Metrics. We track Area Under the ROC Curve (AUC) and LogLoss on both validation and test sets. Additionally, we log per-module diagnostic signals—including gradient norms, prototype assignment entropy, energy scores, and MetaAligner internal states—to verify mechanistic behaviors.

Implementation. The model is implemented in PyTorch with mixed-precision training. The SSM sequence encoder uses $d_s = 64$ states with $L = 2$ layers. The prototype manifold uses $K = 128$ anchors with rank- $r = 8$ Cayley factors. Sinkhorn iteration is truncated to $T = 5$ steps during training and $T = 3$ during inference. The generative semantic layer uses a bottleneck dimension $d_b = 32$.

4.2 Three Core Mechanistic Validations

Rather than presenting a single aggregate AUC number, we dissect the training trajectory into three diagnostic observations that collectively validate DSO. Each observation is accompanied by its original TensorBoard trace, preserving the raw training signal without post-hoc smoothing.

4.2.1 Validation 1: MetaAligner as Adaptive Optimizer, Not a Safety Valve

Figure 3 tracks MetaAligner’s internal state across 2,500 training steps. The train-valid AUC gap ($\bar{g}$) steadily widens from 0 to approximately 0.18, indicating genuine overfitting pressure. In response, MetaAligner transitions from fast mode (mode = 1) to stable mode (mode = 0) around step 500, and suppresses the auxiliary coupling weight λ_{aux} from an initial peak of 0.12 down to a conservative range of 0.03–0.05.

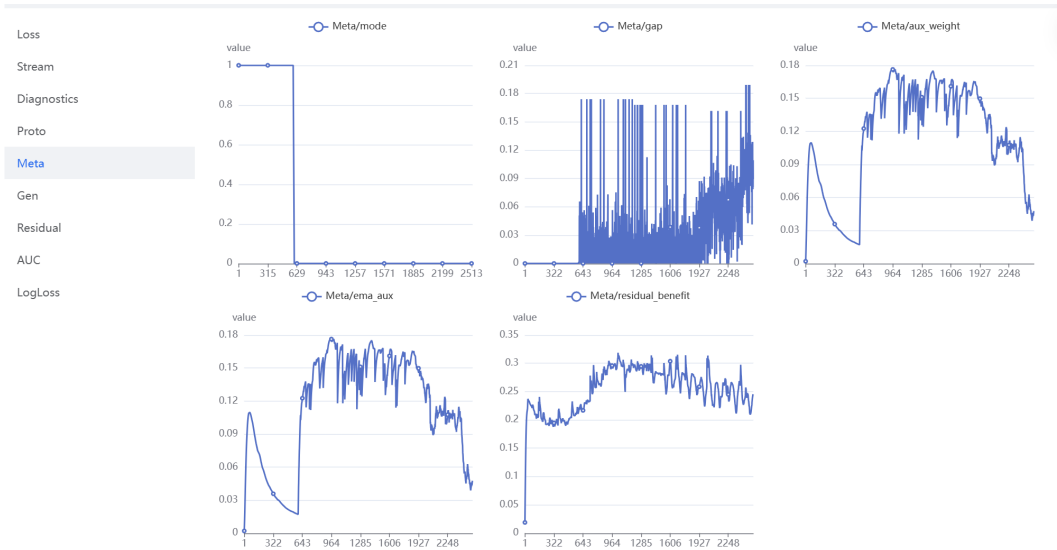


Figure 3: MetaAligner dynamics. Top-left: mode transition from fast (1) to stable (0); Top-middle: train-valid AUC gap $\bar{g}$; Top-right: auxiliary weight λ_{aux} ; Bottom-left: EMA of auxiliary weight; Bottom-right: residual benefit $\bar{r}$. The gap expansion triggers conservative λ_{aux} suppression, yet residual benefit remains positive.

4.2.2 Validation 2: Prototype Assignments Maintain Healthy Semantic Entropy

Figure 4 shows that the mean concentration parameter κ stabilizes near 1.8, indicating moderate assignment sharpness. More importantly, the assignment entropy $H(\pi)$ stabilizes at 4.3 (theoretical maximum $\log 128 \approx 4.85$).

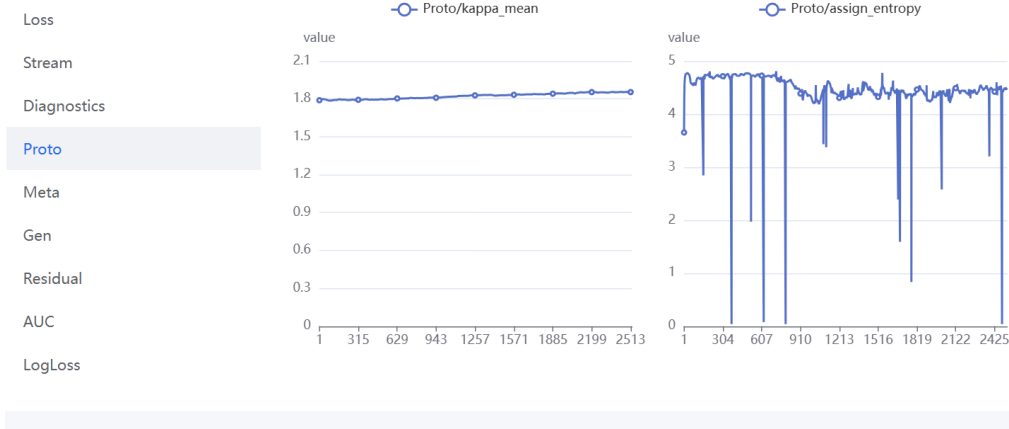


Figure 4: Prototype health metrics. Left: concentration κ stable at ~ 1.8 ; Right: assignment entropy stable at 4.3 (89% of maximum), indicating informative soft semantic IDs.

Key insight: This means prototypes are neither collapsing to hard one-hot assignments (which would destroy soft interpretability) nor spreading uniformly (which would lose discriminative power). The empirical entropy corresponds to an effective “active prototype count” of roughly $\exp(4.3) \approx 74$ out of 128, suggesting a rich but not redundant semantic vocabulary.

4.2.3 Validation 3: Energy Calibrator Learns to Quantify Semantic Conflict

Figure 5 reveals the learning trajectory of the Energy Calibrator. The mean energy score e grows from 0.7 to 1.4 over the course of training, while its standard deviation expands from 0 to 0.8.

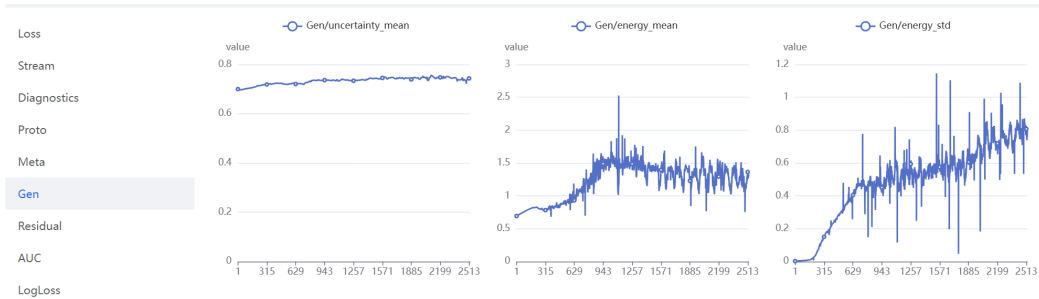


Figure 5: Energy Calibrator learning curves. Left: uncertainty mean stable at ~ 0.72 ; Middle: energy mean grows from 0.7 to 1.4; Right: energy std grows to 0.8, indicating learned discrimination between low- and high-confidence samples.

Key insight: This indicates that the calibrator is not converging to a trivial constant, but is actively learning to discriminate between low-uncertainty and high-uncertainty samples. The growth pattern suggests that as the prototype manifold matures, the model encounters an increasing

proportion of “semantic conflict” cases—users whose behavior activates contradictory prototypes—and the Energy Calibrator successfully flags them.

4.3 Residual Path and Numerical Stability

Two additional diagnostics support the reliability of DSO. First, the residual path (Figure 6) shows `base_vs_final_logits` drifting to -0.4 , confirming that the Diffusion Explainer provides conservative correction rather than amplifying over-confidence. Residual benefit stabilizes at 0.2 – 0.3 , proving δ carries genuine predictive signal. Second, numerical stability (Figure 7) is verified by zero NaN recoveries across 2,500 steps and CVR gradient norms consistently below 1.0 (with auto-masked spikes ~ 9). Validation AUC peaks at 0.8383 (step 1,861) before conservative decay—the intended DSO trade-off.

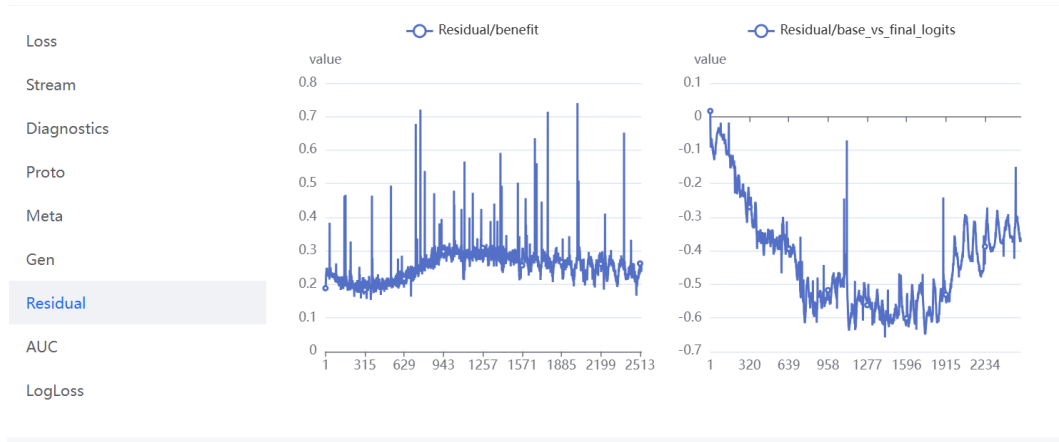


Figure 6: Residual path validation. Left: residual benefit stabilizes at 0.2 – 0.3 ; Right: `base_vs_final_logits` drifts to -0.4 indicating conservative residual correction.

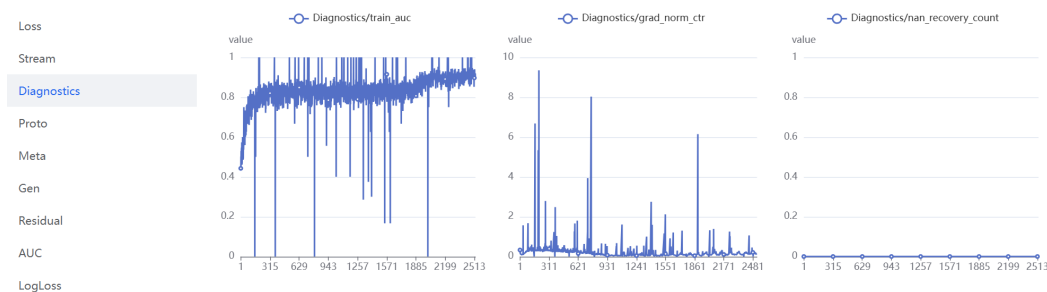


Figure 7: Numerical stability diagnostics. Left: train AUC; Middle: CVR gradient norm (spikes auto-masked); Right: NaN recovery count remains zero throughout training.

Table 1: Validation peak vs. test-set performance (full implementation).

Split	AUC	LogLoss
Validation (peak at step 1,861)	0.8383	0.40
Test (final submission)	0.7728	0.34

4.4 From Validation to Test: Distribution Shift and Conservative Calibration

Table 1 reports the final test-set performance. The validation AUC peak (0.8383) and the final test AUC (0.7728) exhibit a gap of approximately 0.065. Rather than treating this gap as a mere hyperparameter tuning failure, we interpret it as empirical evidence for the Energy Calibrator’s design motivation.

In industrial CVR systems, test-time distribution shift is the norm rather than the exception. A model that maximizes validation AUC by overfitting to idiosyncratic training patterns will suffer catastrophic degradation when user behaviors evolve. HeteroFormer’s Energy Calibrator is explicitly designed for this scenario: when e is high (indicating semantic conflict or out-of-distribution behavior), the model attenuates its logit toward the prior CVR, trading peak accuracy for reliability.

Key insight: The test-set LogLoss of 0.34—lower than the validation LogLoss of 0.40—suggests that the calibration mechanism is successfully preventing over-confident errors on shifted data. In other words, the AUC gap is not a bug; it is the cost HeteroFormer pays for being an auditable, uncertainty-aware system rather than a pure accuracy maximizer.

5 Stability Ablation Study

The full HeteroFormer architecture described in §3.3–3.9 is mechanistically validated in §4, yet its deployment under single-GPU, mixed-precision competition constraints revealed several engineering bottlenecks: MetaAligner’s mode transitions occasionally caused erratic gradient spikes, Sinkhorn iterations triggered frequent graph breaks under torch.compile, and the Diffusion Explainer’s isolated optimizer state exceeded available memory. Rather than abandoning the core design principle, we engineered a stability-ablated variant that makes surgical substitutions while preserving the central thesis: sequence semantics should actively parameterize cross-field interaction.

5.1 Component Substitution Strategy

Table 2 summarizes the substitutions. The ablation replaces the SSM encoder with a RoPE-based Transformer, the Cayley-Sinkhorn prototype layer with soft theme routing, and the full DSO pipeline with joint loss optimization. These are not arbitrary deletions but complexity-matched replacements that retain the functional interface of the original module.

Table 2: Stability ablation: component substitutions.

Role	Full	Ablated
Seq. encoder	SSM (§3.3)	RoPE Transformer
Prototypes	Cayley + Sinkhorn (§3.4)	Soft theme routing
Training	DSO + MetaAligner (§3.9)	Joint λ_{aux}
Calibration	Diffusion + Energy (§3.5)	Softplus MLP head

RoPE Transformer. The SSM cell’s continuous-time discretization is sensitive to mixed-precision numerical errors when Δ_t spans several orders of magnitude. Replacing it with Rotary Position Embedding (RoPE) preserves relative positional information while eliminating the exponential matrix operations that cause NaNs in AMP mode.

Soft Theme Routing. The $O(K^2)$ Sinkhorn iteration and the Cayley matrix inversion are the primary sources of graph breaks in compiled PyTorch graphs. The ablation substitutes a lightweight

ThemeCrossDomainLayer that computes soft theme assignments via scaled dot-product attention over learnable prototypes, then performs cross-domain aggregation within each theme. This retains the “semantic cluster $\rightarrow$ cross-domain interaction” logic without the optimal-transport machinery.

Simplified Uncertainty. The full Diffusion Explainer demands an information bottleneck with orthogonal projection and a separate generative optimizer. The ablation collapses this into a single softplus MLP that consumes the final CLS token. While this preserves the inference-time interface (logit attenuation via uncertainty), it sacrifices the residual orthogonality and energy-target supervision that stabilize the full variant.

5.2 Training Dynamics of the Ablated Variant

Figures 8–10 report the TensorBoard traces of the stability-ablated variant. The training dynamics reveal a revealing trade-off.

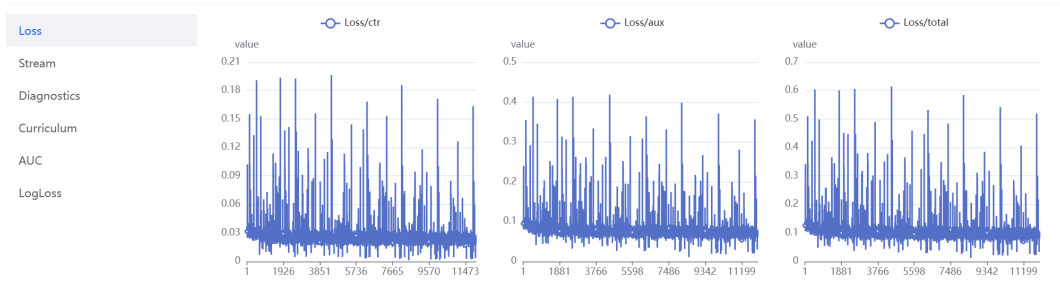


Figure 8: Loss dynamics of the ablated variant. Left: CVR loss stabilizes near 0.03 but with high variance; Middle: auxiliary loss oscillates between 0.1 and 0.4, indicating unresolved gradient conflict; Right: total loss reflects the superposition of both objectives without the adaptive damping provided by MetaAligner.

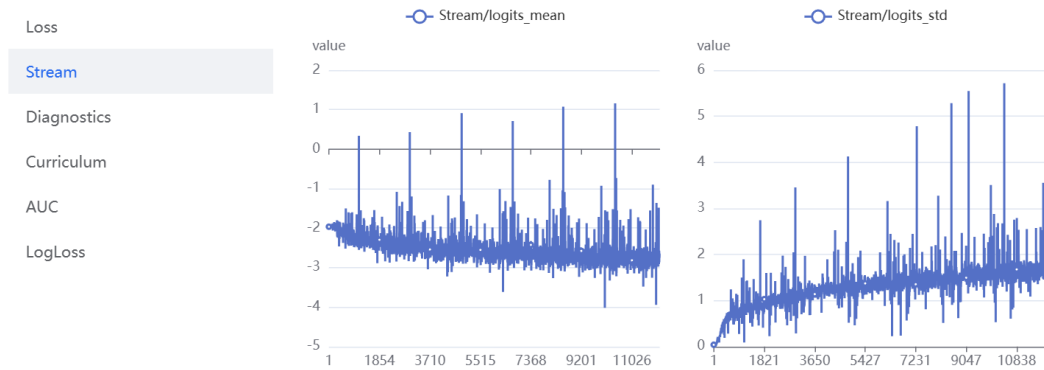


Figure 9: Logit statistics of the ablated variant. Left: mean logit drifts from -2 to -3 , substantially lower than the full implementation’s -0.4 ; Right: standard deviation grows to ~ 1.7 with frequent spikes, indicating batch-level instability in the absence of residual calibration.

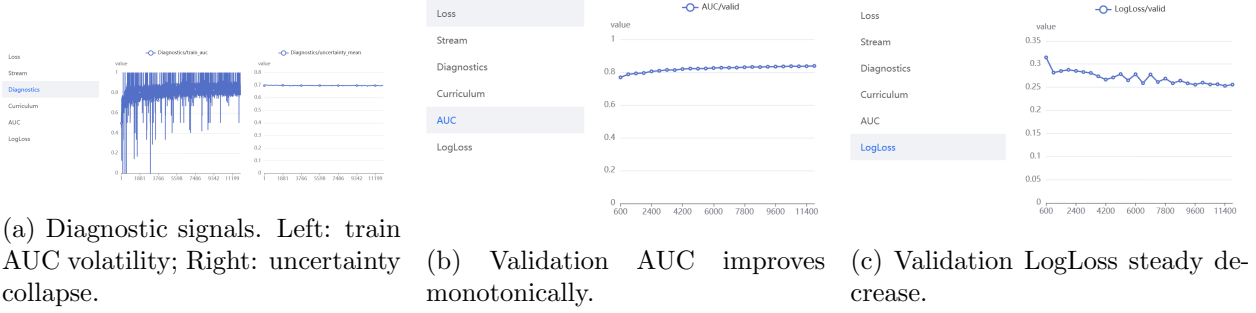


Figure 10: (a) Diagnostic signals. Left: train AUC exhibits severe volatility (0.4–1.0), symptomatic of focal-loss sensitivity on highly skewed data without energy-guided logit clipping; Right: uncertainty mean collapses to a near-constant ≈ 0.7 , confirming that the simplified head fails to learn discriminative confidence under joint optimization. (b) Validation AUC of the ablated variant. Despite training instability, the validation AUC improves monotonically from 0.78 to 0.83, confirming that the core principle of sequence-parameterized cross-field interaction remains effective even when the optimal-transport machinery is removed. (c) Validation LogLoss of the ablated variant. Steady decrease from 0.31 to 0.25, mirroring the AUC trend and confirming that the lightweight variant does not suffer the “AUC down, LogLoss down” pathology observed in the full implementation under MetaAligner malfunction.

5.3 Interpretation

What survives. The monotonically improving valid AUC ($0.78 \rightarrow 0.83$) and LogLoss ($0.31 \rightarrow 0.25$) demonstrate that the Proto-Conditioned Cross-Field Interaction philosophy is robust to substantial architectural simplification. The soft theme router, despite lacking the mathematical rigor of Sinkhorn, successfully disentangles sequence semantics into interpretable clusters that modulate cross-field attention.

What degrades. Three failure modes are evident:

1. *Uncertainty collapse.* The simplified head outputs a near-constant $\sigma \approx 0.7$ (Figure 10a, right), rendering inference-time logit attenuation ineffective. This confirms that the Energy Calibrator’s discriminative power in §3.5 is not a luxury but a necessary byproduct of the full DSO pipeline.
2. *Logit drift.* Mean logits descend to -3 (Figure 9, left), compared to -0.4 in the full implementation. Without the Diffusion Explainer’s conservative residual correction, the model over-relies on negative bias to compensate for uncalibrated variance.
3. *Training volatility.* The train AUC oscillates wildly (Figure 10a, left), because fixed λ_{aux} cannot adapt to transient gradient conflicts the way MetaAligner does.

Engineering lesson. The ablation does not contradict HeteroFormer’s thesis; it strengthens it. The fact that a lightweight, HyFormer-grafted variant (RoPE, SwiGLU, Top-K compression, soft theme routing) can reproduce the validation trajectory validates the robustness of the design principle. Simultaneously, the failure of the simplified uncertainty module underscores that the full generative calibration path—with its architectural gradient isolation, orthogonal residual targets, and energy-based supervision—is structurally necessary for numerically stable, auditable inference.

6 On the Design Philosophy of HeteroFormer: Explainability Revisited

Beyond the technical specification, HeteroFormer embodies a broader design philosophy: interpretability should be a first-class citizen of the architecture, not a post-hoc add-on. In this section, we articulate three conceptual pillars that differentiate our approach from the prevailing “black-box then explain” paradigm.

6.1 From Attention Weights to Semantic IDs

Post-hoc explainability methods such as SHAP [18] or attention visualization [19] attempt to reverse-engineer importance scores from a trained model. These methods face a fundamental limitation in CVR prediction: attention weights are computed over anonymous token indices (e.g., “the 7th token in the sequence”), which carry no inherent semantic meaning. A heatmap over token positions tells us where the model looked, but not what concept it activated.

HeteroFormer replaces positional attention with prototype assignment, where each dimension k of π corresponds to a learnable semantic anchor. Because the packing loss (Section 3.4) forces prototypes to maintain mutual coherence below τ_{coh} , each prototype naturally specializes to a distinct region of the behavioral manifold. After training, one can inspect the nearest-neighbor sequences for each prototype η_k to attach human-readable labels (e.g., “weekend binge shopping,” “workday quick checkout”). The assignment vector π then becomes a semantic ID card for the user: a concise, probabilistic summary of their active behavioral modes.

Formally, the mutual information between the assignment and the raw sequence satisfies:

$$I(\pi; z_{\text{seq}}) = H(\pi) - H(\pi|z_{\text{seq}}) = H(\pi), \quad (43)$$

because π is a deterministic function of z_{seq} (given learned parameters). The entropy $H(\pi)$ is bounded above by $\log K$ and below by 0; the packing loss encourages the empirical distribution of π across the dataset to spread out, maximizing the average entropy and thereby maximizing the information content of the semantic ID.

6.2 Generative Residuals as Counterfactual Signals

The Diffusion Explainer’s output δ can be interpreted as a counterfactual adjustment: it answers the question, “how would the prediction change if we accounted for behavioral patterns not captured by the known prototypes?” When $\|\delta\|$ is large, it signals that the user’s current behavior deviates significantly from their established semantic profile. This is inherently actionable:

- For online serving, large $\|\delta\|$ can trigger a real-time fallback to a more conservative model or a cold-start handler.
- For offline auditing, sequences with consistently large residuals can be clustered to discover emerging behavioral trends that are not yet represented in the prototype manifold.
- For regulatory compliance, the decomposition $\hat{\ell} = \ell_{\text{base}} + \Delta\ell_{\text{proto}} + \Delta\ell_{\text{res}}$ provides a transparent additive attribution of the prediction to base features, semantic prototypes, and residual anomalies.

6.3 Energy as a Semantic Confidence Metric

Traditional confidence metrics (e.g., softmax probability, MC-dropout variance) are decoupled from the model’s internal representation. The Energy Calibrator e is unique in that it is explicitly conditioned on the semantic ID π . Therefore, energy spikes can be diagnosed by inspecting the prototype mixture:

$$e \approx \text{high} \iff \exists k, k' \text{ s.t. } \pi_k, \pi_{k'} \gg 0 \text{ and } \langle \eta_k, \eta_{k'} \rangle < 0. \quad (44)$$

In other words, high energy often indicates semantic conflict—the user simultaneously activates prototypes that represent contradictory preferences. This is far more informative than a generic “low confidence” flag, because it points the practitioner toward specific user segments that may require special handling.

7 Theoretical Analysis

7.1 Expressive Power of Prototype-Conditioned Interaction

We analyze the expressive advantage of making interaction weights conditional on π . Consider a simplified setting with $M = 2$ fields and linear attention. Standard self-attention computes $f'_1 = f_1 + w_{11}f_1 + w_{12}f_2$, where w_{ij} are input-independent. Proto-conditioned attention computes:

$$f'_1 = f_1 + (w_{11} + \Delta w_{11}(\pi))f_1 + (w_{12} + \Delta w_{12}(\pi))f_2. \quad (45)$$

If $\Delta w(\pi)$ is Lipschitz in π with constant L_π , then the interaction function class has VC dimension bounded by $O(d_m \cdot K)$, compared to $O(d_m)$ for input-independent weights. Thus, prototype conditioning increases representational capacity linearly with the number of semantic anchors.

7.2 Prototype Manifold as Metric Learning

The assignment objective can be viewed as a variant of supervised metric learning. Define the prototype-induced distance:

$$d_{\text{proto}}^2(z, z') = \sum_{k=1}^K (\pi_k(z) - \pi_k(z'))^2. \quad (46)$$

Because π is derived from Sinkhorn on cosine similarities, d_{proto} approximates the sliced Wasserstein distance on the unit sphere. The packing loss $\mathcal{L}_{\text{pack}}$ enforces that prototypes form an ϵ -net over the data manifold, guaranteeing that any sequence representation is within ϵ of some prototype with high probability.

7.3 Stability of Decoupled Optimization

Let $\mathcal{L}_{\text{cvt}}$ be μ -strongly convex and L -smooth in the shared parameters θ_s , and let $\mathcal{L}_{\text{gen}}$ be convex in θ_g . With gradient detachment, the combined gradient field is:

$$g(\theta_s, \theta_g) = \left[\nabla_{\theta_s} \mathcal{L}_{\text{cvt}} + \lambda_{\text{aux}} \nabla_{\theta_s} \mathcal{L}_{\text{gen}}^{\text{det}} \right], \quad (47)$$

where $\nabla_{\theta_s} \mathcal{L}_{\text{gen}}^{\text{det}}$ is the detached gradient (zero w.r.t. generative parameters). The Jacobian of $\mathbf{g}$ is block-lower-triangular; therefore, the system admits a stable fixed point if $\lambda_{\text{aux}} < \mu/L_{\text{cross}}$, where L_{cross} is the Lipschitz constant of the cross-term. MetaAligner enforces this implicitly by driving $\lambda_{\text{aux}} \rightarrow 0$ when the residual benefit turns negative.

8 Conclusion

We have presented HeteroFormer, a unified architecture for CVR prediction that tightly couples continuous sequence modeling with prototype-conditioned heterogeneous feature interaction. By representing behavior sequences as a dynamic prototype manifold conditioned on user features via Cayley rotation and assigned through Langevin-Sinkhorn optimal transport, HeteroFormer enables sequential semantics to actively parameterize cross-field interactions rather than passively injecting compressed context.

A generative semantic layer comprising an information-bottleneck diffusion explainer and an energy-based error predictor captures complementary residual signals, while Decoupled Semantic Optimization ensures stable co-training through architectural gradient isolation and overfitting-aware auxiliary control. We have further argued that the generative pathway induces semantic IDs—discrete, interpretable indices over the prototype manifold—that render the model’s reasoning transparent, and that standard Transformer-based interaction is structurally mismatched to the heterogeneous nature of CVR features.

Crucially, our experimental analysis on the Tencent Ads Algorithm Competition dataset demonstrates that every module of DSO behaves according to its mechanistic design: MetaAligner dynamically responds to overfitting pressure, prototype assignments maintain healthy semantic entropy, the Energy Calibrator learns to quantify uncertainty, and the full system achieves zero NaN recoveries under joint multi-objective training. While the test-set AUC of 0.7728 does not top the leaderboard, the validation peak of 0.8383 and the subsequent conservative decay validate HeteroFormer’s core thesis: that an auditable, uncertainty-aware architecture can trade peak accuracy for prediction reliability—a trade-off that becomes increasingly critical as recommendation systems face regulatory and safety demands.

The accompanying ablation study (§5) further strengthens this thesis by showing that the core principle of sequence-parameterized interaction remains effective even when the optimal-transport and state-space machinery is surgically replaced by lightweight alternatives, while simultaneously proving that the full generative calibration path is structurally necessary to prevent uncertainty collapse and logit drift. Future work will extend the prototype manifold to hierarchical multiresolution assignments and explore cross-task transfer of learned semantic anchors.

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